Applied Machine Learning for Plasma Kinetics Simulation

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Low-temperature plasma simulation is a multi-physics problem involving strong coupling among electron kinetics, plasma chemistry, transport, and external operating conditions. One of the major computational bottlenecks in existed *GlobalKin* simulation model is time expensive because it must be repeatedly evaluated as plasma conditions change in time. The goal of this project is to develop a machine-learning-assisted surrogate framework that accelerates plasma kinetics simulation while preserving the essential physics.

For the first stage, the project will train a supervised surrogate for the Boltzmann solver. The input features will include reduced electric field E/N, gas temperature and charged and neutral densities. The output is electron energy distribution function (EEDF) which is high-dimensional structured output. A good approach is to use a latent-space model by compressing the EEDF into a small set of coefficients using principal component analysis (PCA) or an autoencoder. Then a neural network to predict those latent coefficients from the plasma state. In parallel, a multi-output regression network can be trained to predict the rate coefficients directly. Feed Forward Network with residual connections is likely the best initial model. To improve physical consistency, the loss function can include penalties that enforce non-negativity, smoothness, and agreement between EEDF-derived and directly predicted rate coefficients.

For the second stage, the project will build a surrogate for the time-dependent 0D plasma kinetics simulation. A second learned model can be trained to approximate the time evolution of the plasma state using a Neural ODE or recurrent neural network. Not only using the time-independent input such as power, flow rate, frequency, volume etc, but also the surrogate model for Boltzmann solver developed in the first state will be used to determine EEDF and the corresponding rate coefficients of chemistry in low temperature plasmas. Model performance will be evaluated using error metrics on EEDF shape, rate coefficients, and time-dependent species densities, as well as computational speedup relative to the original solver.

The dataset includes the densities of species, electron temperature, and energy deposition, EEDF. The raw simulation outputs from 0D model *GlobalKin* will be cleaned and aligned so that each sample contains a consistent set of input plasma-state variables and corresponding outputs. Selected variables e.g. EEDF will be transformed using logarithmic scaling to improve numerical stability and learning performance. All input and output features will then be normalized or standardized before training. Jisu is mainly responsible for the data preparation and pre/postprocessing. Bingqing will work on the machine learning model development and parameter tuning.

If the fully integrated machine learning surrogate for the coupled Boltzmann–ODE system does not achieve sufficient accuracy, the project will focus on applying machine learning only to the Boltzmann solver to accelerate the most computationally expensive part of the simulation. In addition, techniques beyond those covered in the course, such as physics-informed neural networks may be explored if they improve predictive accuracy, physical consistency, or scalability.

Appendix

The dataset could be download via
https://github.com/bqwangumich/NERS_590_Final_Project/blob/main/NERS590_electron_argon_data.csv

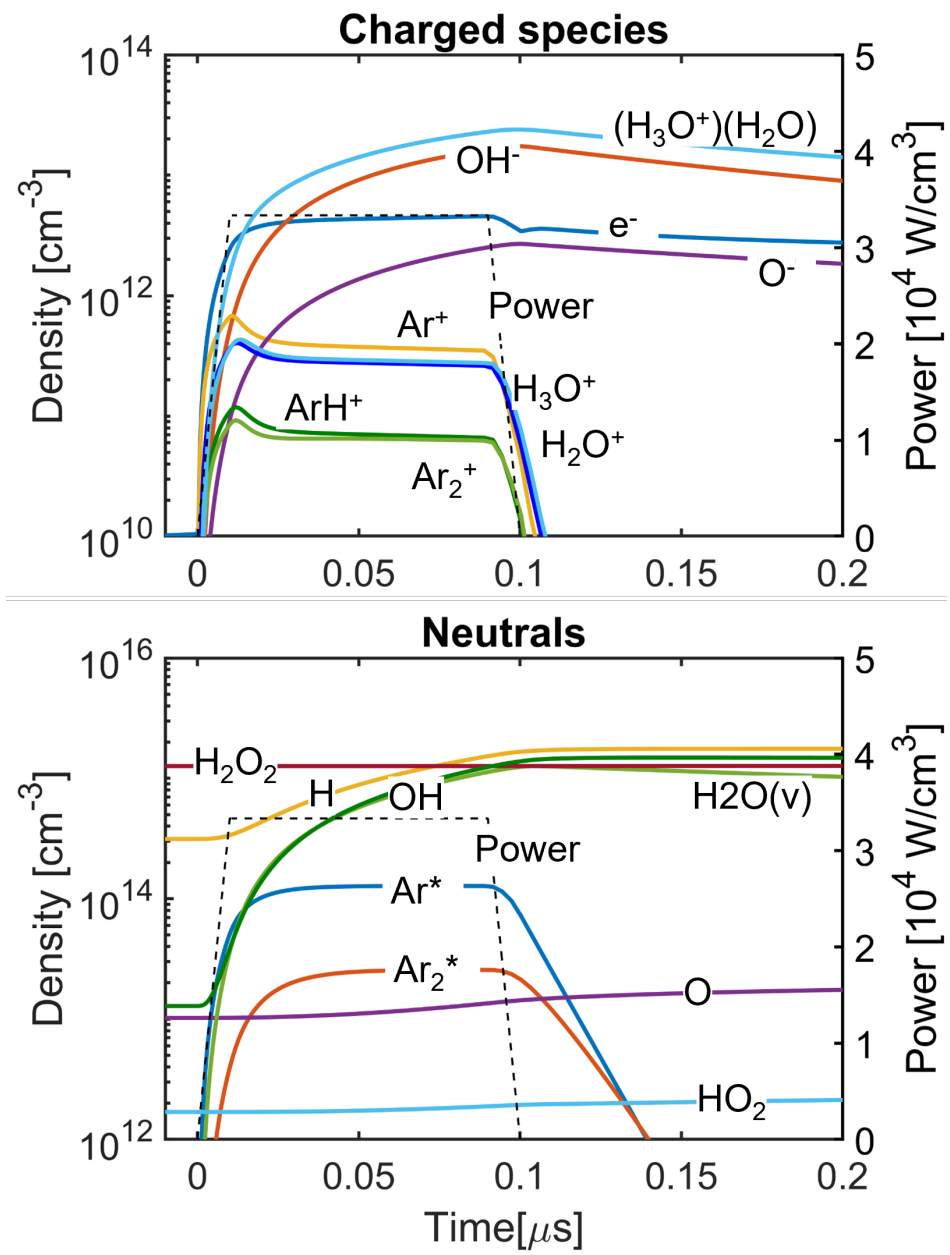

Figure 1. Charged and neutral species in 0D DBD discharge